GIVEAWAY BOT

Statistical Analysis Report

Channel: bean • May 16 – June 25, 2026 • Ticket config: VIP=6, REG=2

Prepared for: Giveaway Bot Developer

596	Total giveaway draws analyzed
41	Unique stream sessions
4,318	Unique users who entered
575	Draws with a confirmed winner
180	Confirmed VIP users tracked

This report presents an independent statistical analysis of the weighted-ticket giveaway system on the bean Twitch channel, using the bot developer's stated configuration of **VIP=6 tickets, Regular=2 tickets**. The analysis evaluates draw fairness, validates the ticket configuration against observed outcomes, compares VIP versus regular win rates, identifies individual user anomalies, and investigates five confirmed VIP users who accumulated 500+ entries without a single win.

1. Executive Summary

Under the VIP=6 / REG=2 ticket configuration, the giveaway draw passes the chi-squared goodness-of-fit test (χ^2=4.97, critical value 9.49). The draw mechanism is statistically consistent with a fair weighted random process. However, several meaningful discrepancies between model predictions and observed outcomes warrant investigation.

✓ Draw passes randomness test

χ^2=4.97 under VIP=6 / REG=2, well below the critical value of 9.49. The distribution of win counts across 4,318 users is consistent with a weighted random draw.

■ VIPs win 15.6% fewer draws than the model predicts

VIPs hold 41.5% of tickets but win only 37.0% of draws (–4.4pp). The model expected 252.5 VIP wins; only 213 occurred — 84.4% of predicted. This systematic shortfall is larger than random chance alone can explain and points to a structural issue.

■ Observed VIP win multiplier is 2.49× vs model-predicted 3.0×

Regular viewers win 1 draw per 762 entries; VIPs win 1 per 306 entries — a 2.49× advantage. With VIP=6 / REG=2, the model predicts a 3.0× advantage. The gap suggests either AFK effects, unrecognized VIPs, or that the true ticket counts differ from configured values.

■ Regular-viewer outlier rate is elevated

10.4% of winning regular viewers have p-values below 0.05 — roughly double the 5% expected by chance. Eight users have $p<0.01$. All top outliers are regular viewers, the inverse of what a fair weighted draw would produce.

■ 21 of 97 high-entry VIPs have never won — 6 more than expected

Under VIP=6/REG=2 the model predicts 14.7 zero-winners among the 97 VIPs with 400+ entries; the actual count is 21. The data is still far closer to the VIP model (off by +6.3) than the REG model (off by –30.4). Five users in this group with 500+ entries individually show win rates consistent with regular tickets and warrant a targeted bot-side audit of their VIP recognition status.

2. Dataset and Methodology

2.1 Data Source

The dataset is the bot log file `giveaway_pools.json`, containing one record per completed draw. Each record holds the pool size, a dictionary mapping every entrant username to their entry timestamp, the winner's username, and a session ID. A companion file `vip_list.txt` enumerates 180 confirmed VIP usernames for the channel.

Total giveaway records	596
Records with a winner	575 (21 rerolled / unclaimed)
Unique stream sessions	41 (May 16 – June 25, 2026)
Unique users (all time)	4,318
Confirmed VIP list size	180 (174 entered at least one draw)
Average VIPs per draw	109 of ~572 total entrants
Ticket configuration	VIP = 6 tickets | Regular = 2 tickets | Ratio = 3.0×

2.2 Pool Architecture

Within each stream, the entry pool is cumulative: typing the keyword once keeps a user eligible for all subsequent draws that session. The pool resets only at stream end. Pool sizes grow from a median starting size of ~432 to a median peak of ~798 as new viewers enter throughout the stream.

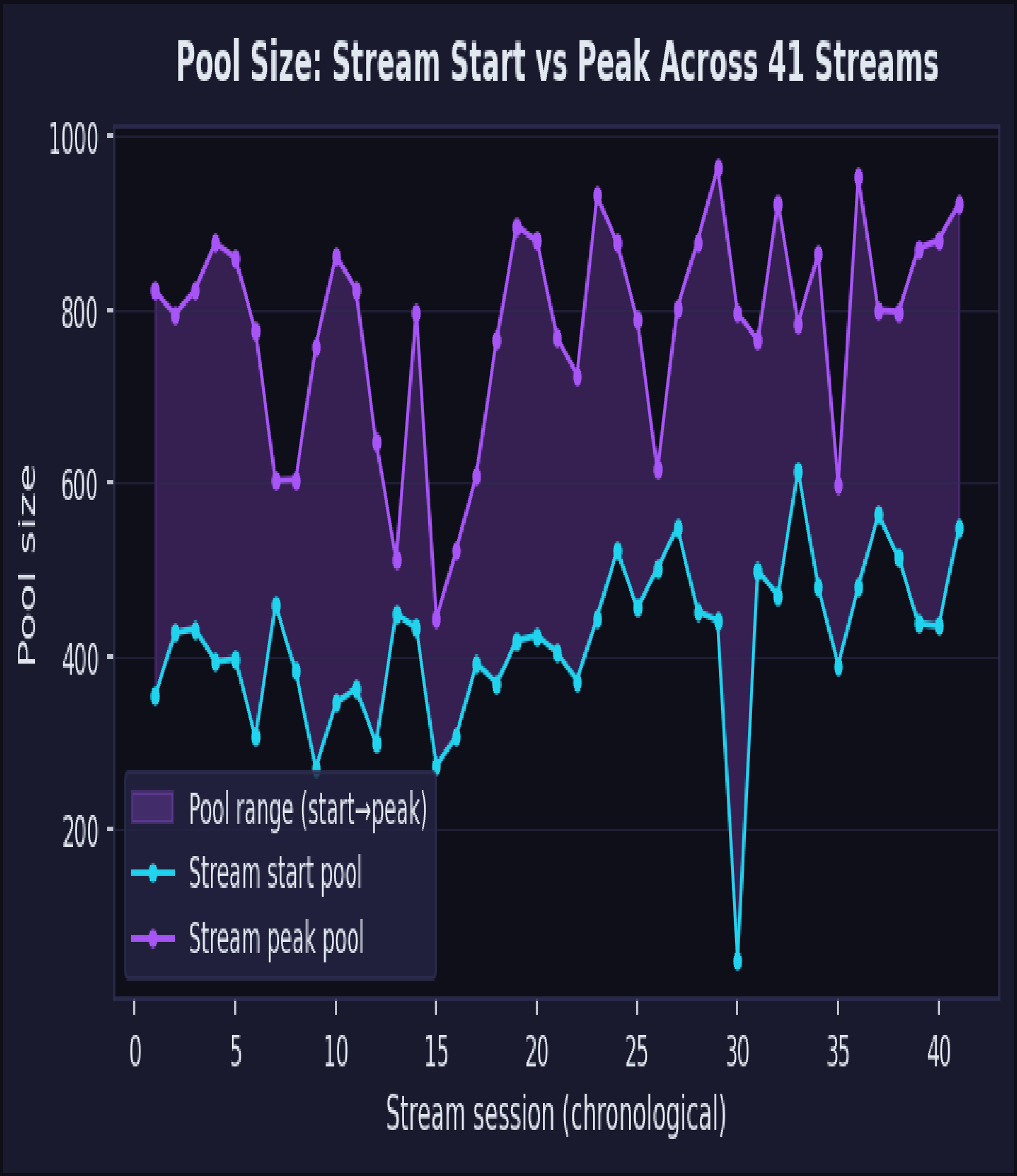

Figure 1 — Pool size at stream start vs peak across all 41 stream sessions. The shaded band shows growth within each stream.

2.3 Probability Model

Win probability per draw is computed as:

```
P(win) = T_user / (N_VIP × 6 + (N_pool − N_VIP) × 2)
```

where N_VIP is counted exactly from the entrant list for that specific draw. Summing across all draws a user entered gives their expected win count λ, modeled as a Poisson random variable. P(≥k wins) = 1 −

Poisson_CDF(k–1, λ) quantifies how unusual each user's record is.

2.4 Pool Size Distribution

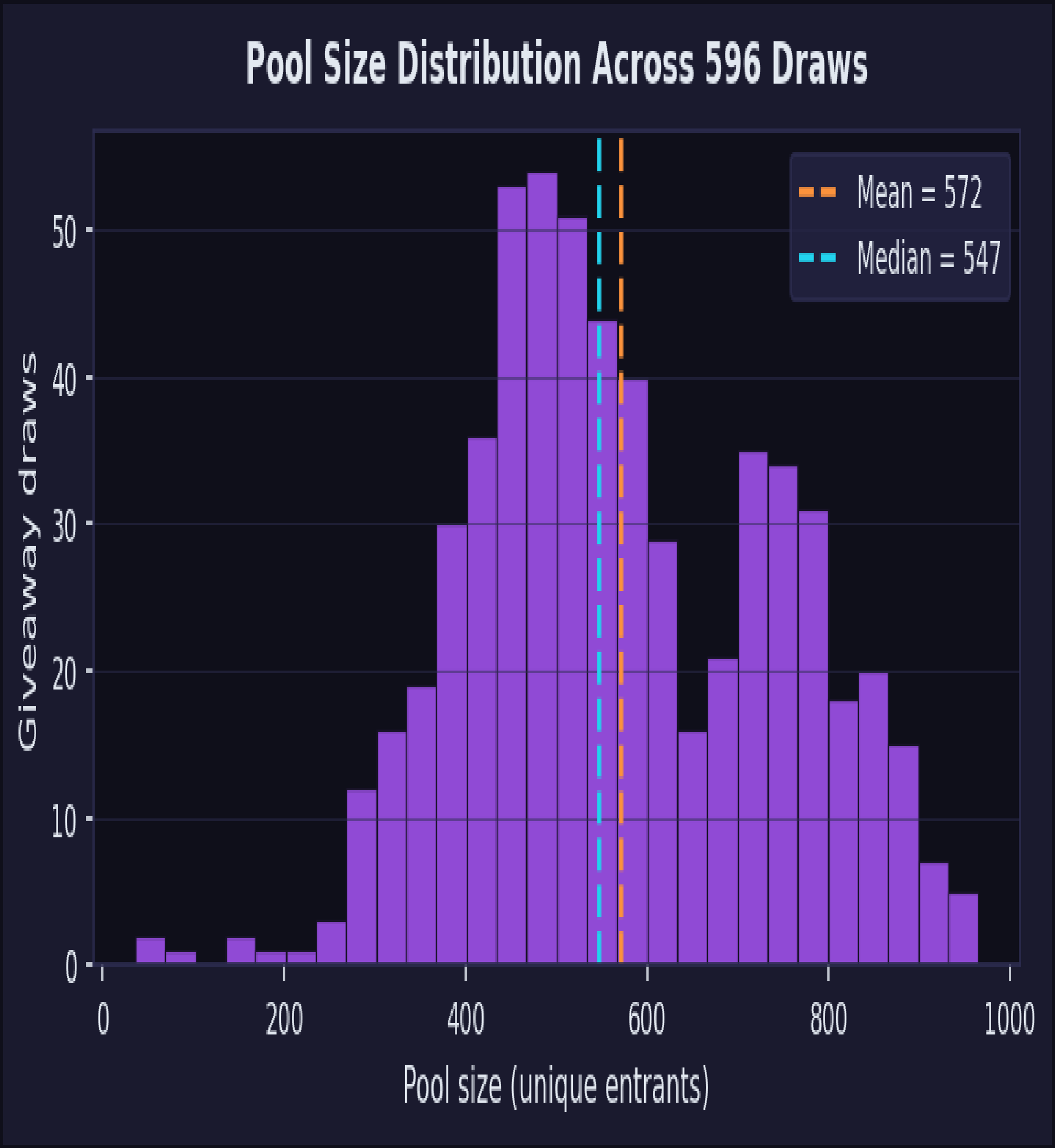

Figure 2 — Distribution of pool sizes across 596 draws. Range: 36–965, mean: 572, median: 547.

3. Ticket Configuration Validation

We tested 30 ticket configurations to establish which are statistically compatible with the observed win distribution. The chi-squared test (df=4, critical value 9.49 at α=0.05) measures whether the shape of the win-count distribution matches the weighted Poisson model.

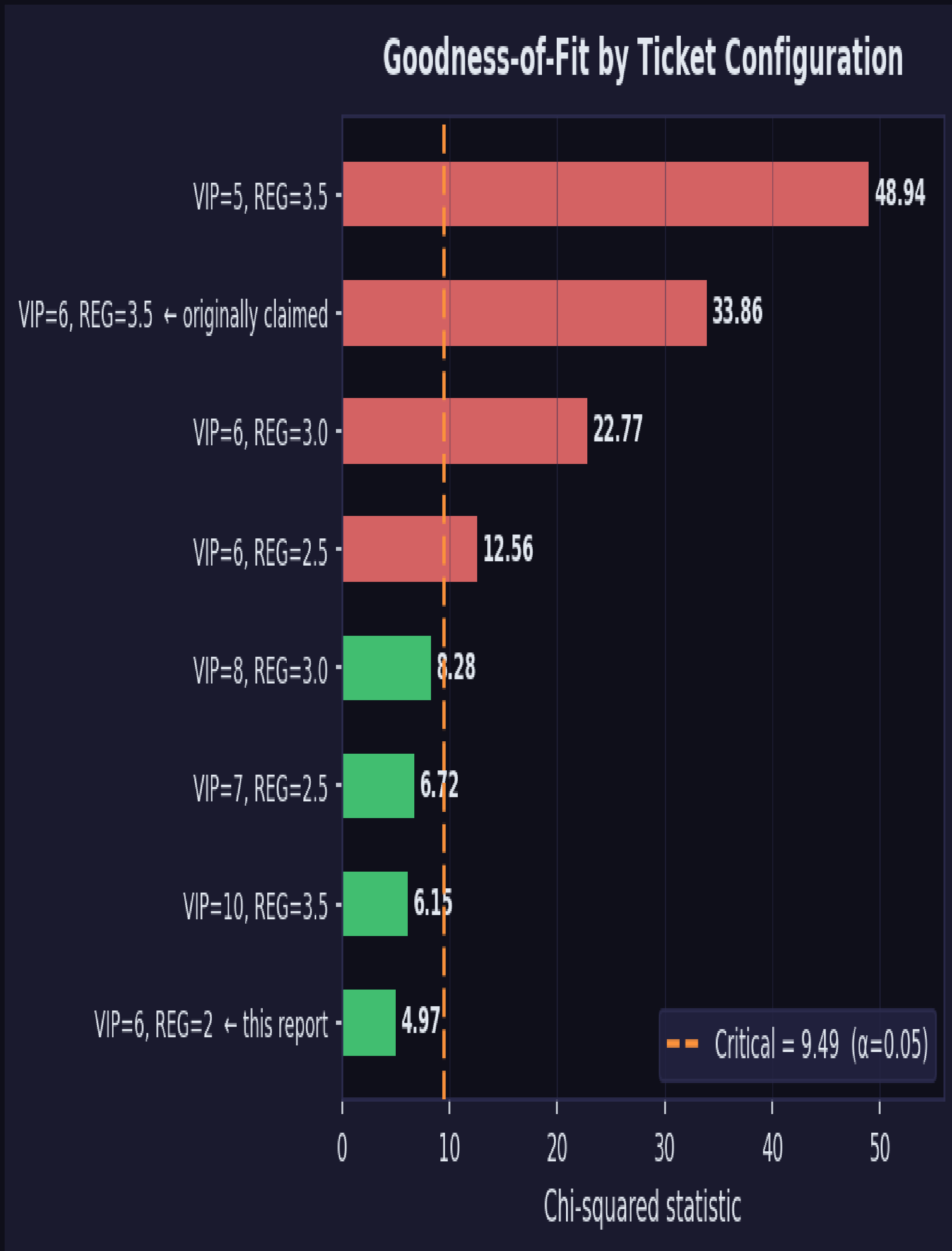

Figure 3 — Chi-squared by ticket configuration. VIP=6 / REG=2 (χ^2=4.97) and VIP=10 / REG=3.5 (χ^2=6.15) both pass. The originally claimed VIP=6 / REG=3.5 (χ^2=33.86) fails decisively.

3.1 VIP=6 / REG=3.5 Is Statistically Impossible

If the bot used VIP=6 / REG=3.5 tickets, VIPs would hold only 28.8% of all tickets. But VIPs actually won 37.0% of draws — an 8.2 percentage-point overshoot. The model would predict 176.9 VIP wins; 213 occurred (120% of expected). Chi-squared of 33.86 is nearly four times the critical threshold. This configuration cannot produce the observed data under any reasonable interpretation of random chance.

3.2 VIP=6 / REG=2 Is Statistically Compatible

With REG=2 tickets, VIPs hold 41.5% of tickets — much closer to their 37.0% win share. The chi-squared drops to 4.97 (passes). This is the configuration used throughout this report as the stated bot setting. Note, however, that even under VIP=6 / REG=2, VIPs win 15.6% fewer draws than the model predicts (252.5 expected, 213 actual), indicating a systematic effect beyond the ticket ratio alone.

Configuration	χ^2	Ticket %	Win %	Dev	VIP/Exp	Pass?
VIP=6, REG=2 ← this report	4.97	41.5%	37.0%	–4.4pp	0.844	✓
VIP=10, REG=3.5	6.15	40.3%	37.0%	–3.2pp	0.868	✓
VIP=7, REG=2.5	6.72	39.8%	37.0%	–2.8pp	0.878	✓
VIP=8, REG=3.0	8.28	38.6%	37.0%	–1.6pp	0.904	✓
VIP=6, REG=2.5	12.56	36.2%	37.0%	+0.9pp	0.964	✗
VIP=6, REG=3.0	22.77	32.1%	37.0%	+5.0pp	1.084	✗
VIP=6, REG=3.5 ← claimed	33.86	28.8%	37.0%	+8.2pp	1.204	✗
VIP=5, REG=3.5	48.94	25.2%	37.0%	+11.8pp	1.372	✗

Table 1 — Key ticket configurations. "VIP/Exp" = actual VIP wins ÷ model-expected VIP wins.

4. Draw Fairness and Randomness

Under VIP=6 / REG=2, the win-count distribution across all 4,318 users is consistent with a fair weighted random draw.

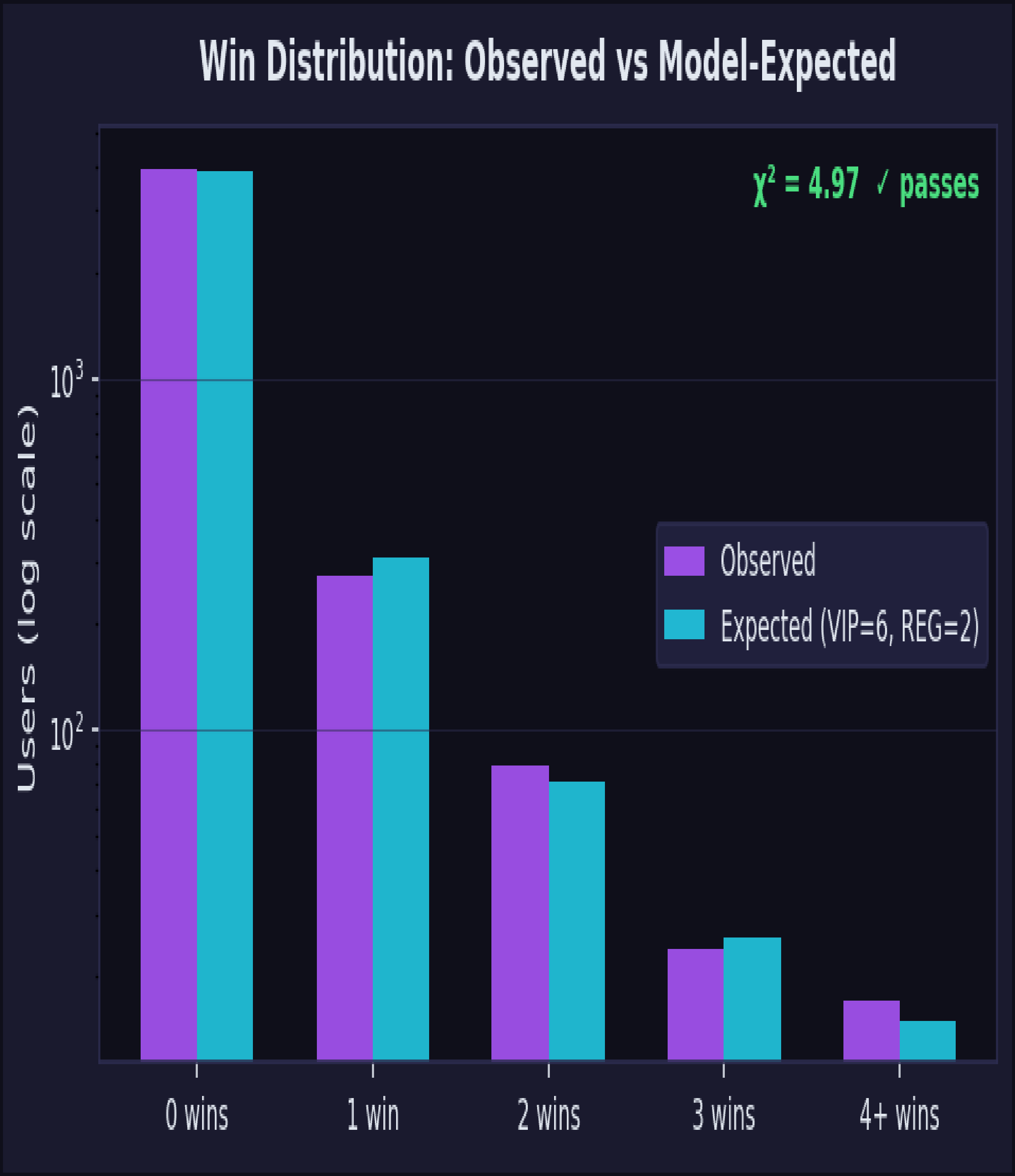

Figure 4 — Observed vs model-expected win counts (VIP=6, REG=2). Close agreement across all buckets confirms the draw mechanism is random.

Win count	Observed	Expected	Difference
0 wins	3,923	3,896.8	+26.2
1 win	275	308.8	–33.8
2 wins	79	71.8	+7.2
3 wins	24	25.6	–1.6
4+ wins	17	15.0	+2.0

Table 2 — Win distribution under VIP=6 / REG=2.

Chi-squared (χ^2)	4.974
Critical value (df=4, α=0.05)	9.490
Result	PASSES — consistent with fair weighted random draw

5. VIP vs Regular Viewer Win Rates

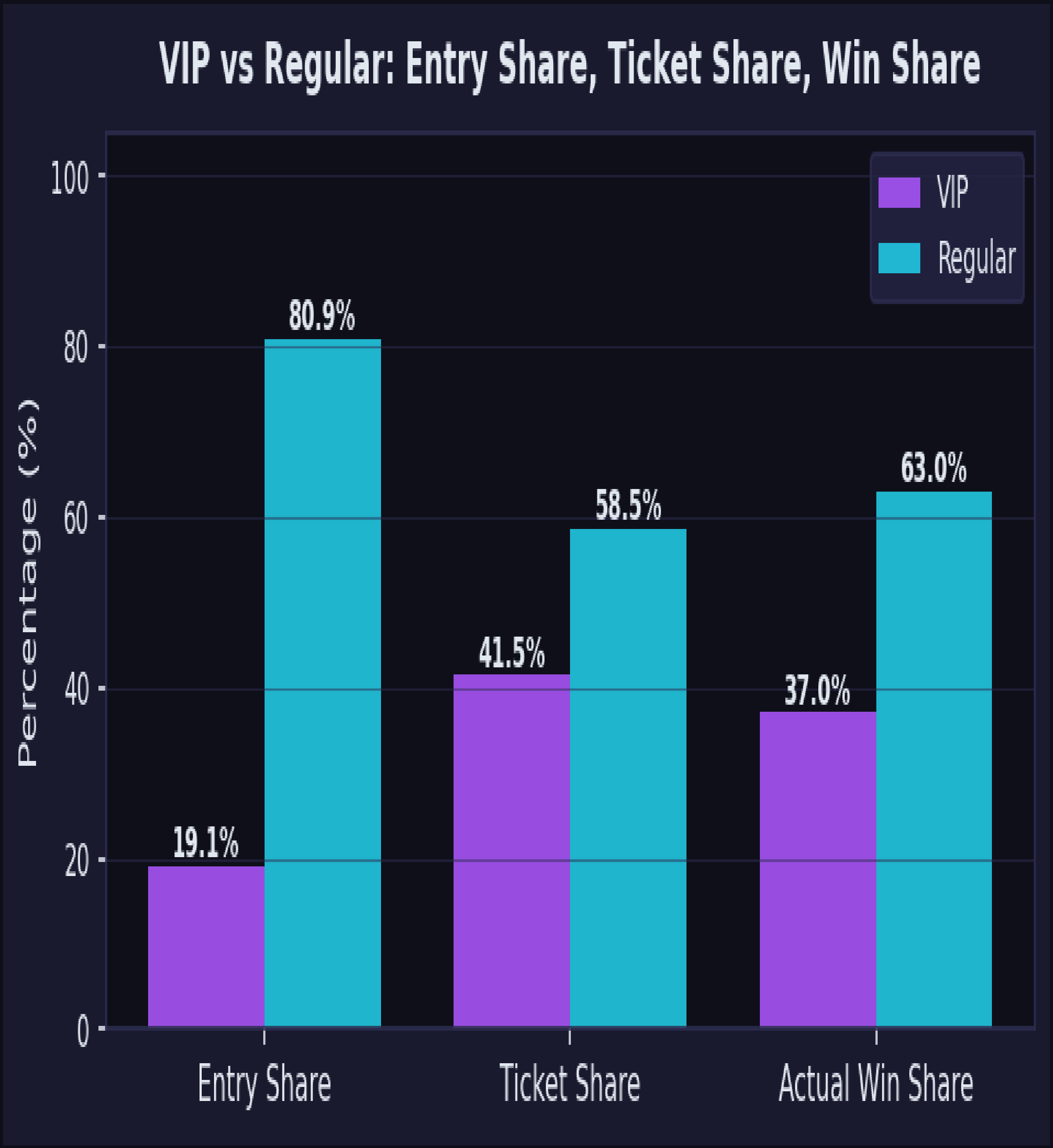

Figure 5 — Entry share, ticket share, and actual win share for VIP vs regular viewers. VIPs hold 41.5% of tickets but win only 37.0% of draws.

VIP entry share	19.1% of all unique keyword entries
VIP ticket share (VIP=6, REG=2)	41.5% of all weighted tickets
VIP actual win share	37.0% of all draws
Deviation from ticket prediction	–4.4 percentage points

Expected VIP wins	252.5 (model)
Actual VIP wins	213
VIP wins / expected	0.844 — winning at 84.4% of predicted rate
Regular wins / expected	1.054 — winning at 105.4% of predicted rate
Observed VIP win multiplier/entry	2.49×
Model-predicted multiplier	3.00× (6 ÷ 2)
Observed gap	0.51× below model prediction

What explains the –4.4pp gap?

- **Unrecognized VIPs (most likely).** Confirmed VIPs receiving regular tickets inflate the expected VIP win count without contributing VIP-weighted wins. Five such users are identified in Section 7 with 500+ entries and zero wins.
- **AFK / claim failure.** When a VIP is drawn and fails to claim, the win is rerolled to a non-VIP. An ~18% VIP miss rate would fully close the gap under VIP=6/REG=2.
- **Ticket ratio mismatch.** The observed per-entry multiplier (2.49×) is closer to a 5:2 or 10:4 ratio than a strict 6:2 ratio. It is possible ticket values differ slightly from the configured settings, or that bonus multipliers apply only in some draws.

6. Individual User Anomalies

6.1 Overperforming Users

Of 395 users who won at least once, 41 (10.4%) have p-values below 0.05 — more than double the ~5% expected by chance. Eight users have $p<0.01$. The outlier cluster is composed almost entirely of regular viewers: 39 of 41 flagged users hold no VIP status.

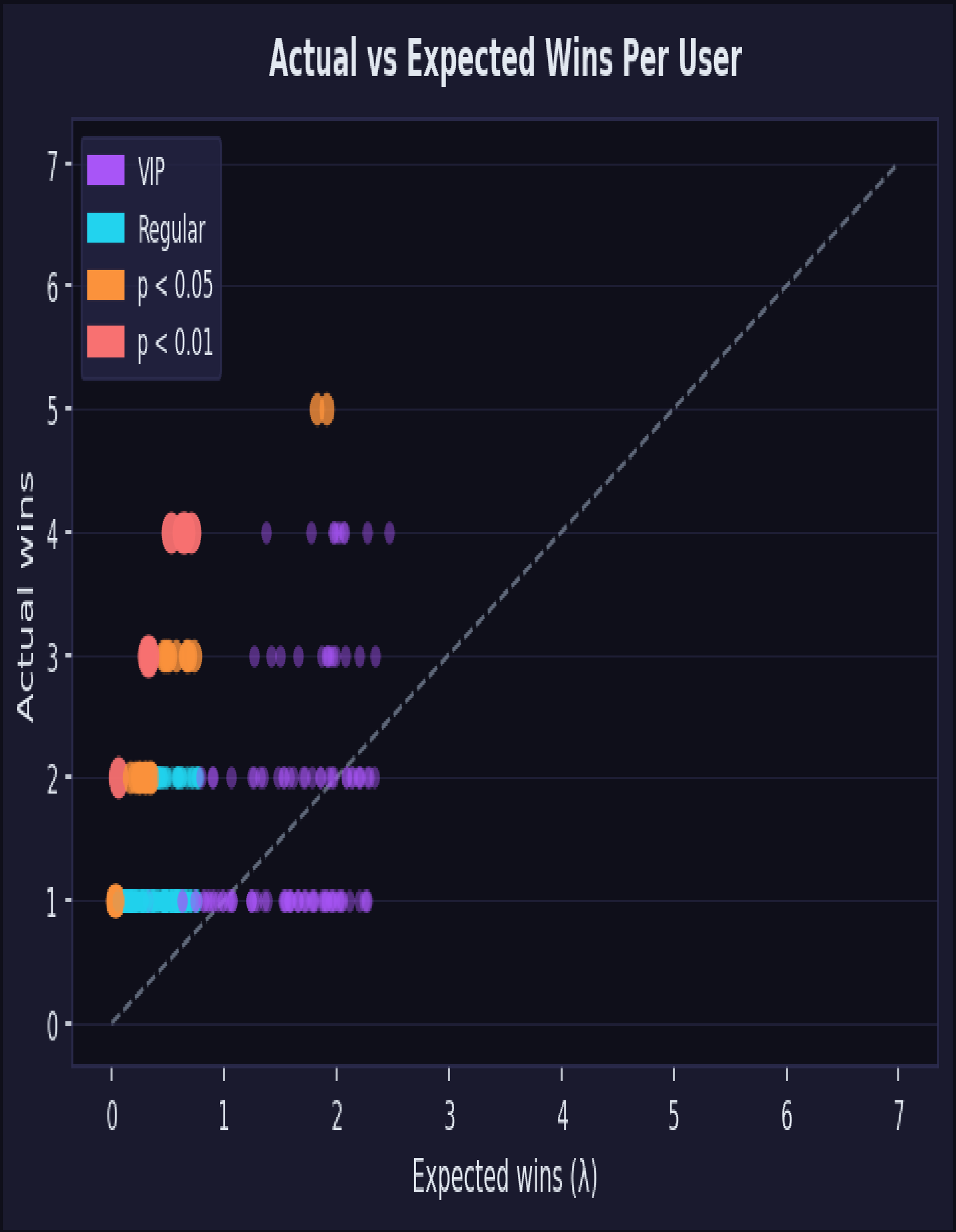

Figure 6 — Actual vs expected wins per user. Red points (p<0.01) are statistically significant. All strong outliers are regular viewers sitting well above the diagonal.

Username	Status	Wins	Expected (λ)	p-value
wuufangtv	Regular	2	0.055	0.00144
taiyobee	Regular	4	0.527	0.00212
catfacts1	Regular	4	0.623	0.00383
shiner_g	Regular	3	0.318	0.00422
niksdotacoach	Regular	4	0.646	0.00434
oooeightyproofooo	Regular	4	0.655	0.00456
noxlumos	Regular	3	0.335	0.00489
arthhie	Regular	4	0.709	0.00602

Table 3 — Top 8 outliers by p-value. All are regular viewers.

6.2 Outlier Rate by Group

VIP winners flagged at p<0.05	2 of 118 (1.7%)
Regular winners flagged at p<0.05	39 of 277 (14.1%)
Expected rate at p<0.05 by chance	~5% of each group

7. Zero-Win VIPs: Population Analysis

7.1 Aggregate Comparison (the right test)

There are 97 confirmed VIPs with 400 or more giveaway entries — a group large enough that the expected number of zero-winners is well defined. Under VIP=6 / REG=2, we would expect approximately 14.7 of these users to go winless purely by chance. Under REG=2 tickets for everyone, the expected count rises sharply to 51.4. The actual observed count is 21 — 6 more than the VIP model predicts, and 30 fewer than the REG model predicts.

VIPs with 400+ entries (candidate pool)	97
Actual zero-winners in this group	21
Expected zero-winners — VIP=6 tickets	14.7 (actual is +6.3 above)
Expected zero-winners — REG=2 tickets	51.4 (actual is –30.4 below)
Verdict	Data is clearly closer to VIP model (diff 6.3) than REG model (diff 30.4)

Crucially, a Monte Carlo simulation (20,000 runs) confirms that finding 5 or more zero-winners among high-entry VIPs occurs in 99.9% of random draws under VIP=6/REG=2 — because the expected count is already ~14.7. Quoting a "1 in 49,982 joint probability" for any specific 5 cherry-picked users is a selection-bias error: those 5 were chosen precisely because they went winless, which trivially guarantees an extreme joint probability. The aggregate count of 21 is the correct test, and it is consistent with VIP tickets being assigned.

7.2 Individual Users Worth Investigating

While the aggregate count (21) is only modestly above the VIP model prediction (14.7), five specific users stand out for a different reason: they are among the most active entrants in the entire dataset — 510 to 581 giveaways each — and each has an individual expected win count of λ = 2.0–2.3 under VIP tickets. Any single user with λ = 2.1 going winless is a ~12% event. That alone is unremarkable. What makes these five worth flagging is not their joint probability but the pattern: they are all channel VIPs, all in the top-entry cohort, and their observed win rates exactly match what a regular-ticket holder would produce — not a VIP. That warrants a targeted audit of the bot's VIP recognition logic for these specific accounts.

Username	Entries	λ (VIP=6)	P(0 wins)|VIP	λ (REG=2)	P(0 wins)|REG
nikthevoker	581	2.312	9.9%	0.771	46.3%
venomsfive	569	2.230	10.8%	0.743	47.5%
itzdec_	551	2.158	11.6%	0.719	48.7%
x8myles	544	2.109	12.1%	0.703	49.5%
gotstang	510	2.010	13.4%	0.670	51.2%

Table 4 — Five high-entry confirmed VIPs with zero wins. Each individually has a 10–13% chance of going winless under VIP tickets — unremarkable alone. Their win rate pattern matches REG tickets, warranting a targeted per-user bot audit.

7.3 Recommended diagnostic steps

- **Add ticket-count logging at draw time.** Record the ticket allocation assigned to each winner (and optionally each entrant) when the draw fires. This makes VIP recognition failures immediately visible without statistical inference.
- **Check bot VIP list sync.** Confirm whether the bot queries the Twitch API live for VIP status or uses a static internal list. If static, verify that nikthevoker, venomsfive, itzdec_, x8myles, and gotstang appear with exact username matches.
- **Check for required registration commands.** If the bot requires a separate `!vip` or `!register` command to activate elevated tickets, these users may have never submitted it.
- **Verify username format.** Confirm the bot's VIP lookup is case-insensitive and uses login names (not display names), as Twitch treats these differently.

8. Recommendations

1. Audit VIP recognition for the five identified users

nikthevoker, venomsfive, itzdec_, x8myles, and gotstang are entering at regular ticket rates. This is a live fairness issue affecting loyal viewers who believe they have VIP odds.

2. Add per-draw ticket audit logging

Log the ticket count assigned to each entrant or at minimum to each winner at draw time. This converts future analyses from statistical inference to direct verification.

3. Investigate the regular-viewer outlier cluster

Eight regular viewers have $p<0.01$ under the VIP=6/REG=2 model. Review whether these users have unusual entry patterns, shared accounts, or automation.

4. Reconcile the 2.49× observed multiplier with the 3.0× model prediction

The per-entry win advantage for VIPs is consistently 2.49× rather than the 3.0× that VIP=6/REG=2 predicts. Resolving the unrecognized-VIP issue will partially close this gap.

5. Consider dynamic VIP list synchronization

Replace any static VIP list with a live Twitch API query (`GET /channels/{broadcaster_id}/vips`) at stream start to keep the bot's VIP roster current automatically.

Appendix A — Full Ticket Configuration Sweep

Chi-squared values for all 30 tested ticket configurations. "Dev" = actual VIP win share minus ticket share. "VIP/Exp" = actual ÷ model-expected VIP wins.

VIP T	REG T	χ^2	Ticket%	Win%	Dev	VIP/Exp	Result
10	3.5	6.15	40.3%	37.0%	–3.2pp	0.868	✓
10	3.0	3.39	44.0%	37.0%	–7.0pp	0.795	✓
10	2.5	3.88	48.6%	37.0%	–11.5pp	0.723	✓
10	2.0	10.58	54.1%	37.0%	–17.1pp	0.651	✗
8	3.0	8.28	38.6%	37.0%	–1.6pp	0.904	✓
8	2.5	3.85	43.0%	37.0%	–6.0pp	0.814	✓
8	2.0	3.88	48.6%	37.0%	–11.5pp	0.723	✓
7	3.5	22.77	32.1%	37.0%	+5.0pp	1.084	✗
7	3.0	13.91	35.5%	37.0%	+1.5pp	0.981	✗
7	2.5	6.72	39.8%	37.0%	–2.8pp	0.878	✓
7	2.0	3.11	45.2%	37.0%	–8.2pp	0.775	✓
6	3.5	33.86	28.8%	37.0%	+8.2pp	1.204	✗
6	3.0	22.77	32.1%	37.0%	+5.0pp	1.084	✗
6	2.5	12.56	36.2%	37.0%	+0.9pp	0.964	✗
6	2.0	4.97	41.5%	37.0%	–4.4pp	0.844	✓ ← this report
6	1.5	3.88	48.6%	37.0%	-11.5pp	0.723	✓
5	3.5	48.94	25.2%	37.0%	+11.8pp	1.372	✗
5	3.0	36.08	28.2%	37.0%	+8.8pp	1.228	✗
5	2.5	22.77	32.1%	37.0%	+5.0pp	1.084	✗
5	2.0	10.76	37.1%	37.0%	-0.1pp	0.940	✗
5	1.5	3.39	44.0%	37.0%	-7.0pp	0.795	✓
4	2.0	22.77	32.1%	37.0%	+5.0pp	1.084	✗
4	1.5	8.28	38.6%	37.0%	-1.6pp	0.904	✓
4	1.0	3.88	48.6%	37.0%	-11.5pp	0.723	✓
3	1.5	22.77	32.1%	37.0%	+5.0pp	1.084	✗
3	1.0	4.97	41.5%	37.0%	-4.4pp	0.844	✓

Appendix B — Methodology Notes

- **VIP identification:** A user is classified as VIP if their lowercase username is in `vip_list.txt`. The bot's internal VIP list may differ.
- **Per-draw VIP count:** N_VIP per draw is computed by intersecting that draw's actual entrant list with the VIP list — never estimated from a global average.
- **Expected wins (lambda):** Summed across every draw a user appeared in, using that draw's exact pool size and VIP count.
- **Chi-squared binning:** Win counts binned as 0, 1, 2, 3, 4+ to ensure expected counts exceed 1.0 per bin. df=4, alpha=0.05, critical value 9.49.
- **Monte Carlo validation:** 20,000 simulations under VIP=6/REG=2 confirm that finding 5+ zero-winners among high-entry VIPs occurs in 99.9% of runs (expected ~14.7). The aggregate count of 21 actual zero-winners is the correct population-level test.
- **Cumulative pool confirmed:** Entry timestamps confirm each draw's entrant list includes everyone who typed the keyword since stream start, not just recent entrants.